

# Supplementary Materials

Table 1: Classification Tasks and Corresponding SOTA References across Modalities.

| Modality | Ref    |
|----------|--------|
| Images   | [5, 6] |
| Video    | [4]    |
| MTS      | [3]    |
| Audio    | [1]    |

Table 2: Regression and Forecasting Tasks with SOTA References across Modalities.

| Modality | Ref |
|----------|-----|
| Image    | [5] |
| Video    | [4] |
| Audio    | [2] |
| MTS      | [3] |

## References

- [1] Ke Chen, Xingjian Du, Bilei Zhu, Zejun Ma, Taylor Berg-Kirkpatrick, and Shlomo Dubnov. Hts-at: A hierarchical token-semantic audio transformer for sound classification and detection. In *ICASSP 2022*.
- [2] Ke Chen, Xingjian Du, Bilei Zhu, Zejun Ma, Taylor Berg-Kirkpatrick, and Shlomo Dubnov. Hts-at: A hierarchical token-semantic audio transformer for sound classification and detection. In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 646–650. IEEE, 2022.
- [3] Xiwen Chen, Peijie Qiu, Wenhui Zhu, Huayu Li, Hao Wang, Aristeidis Sotiras, Yalin Wang, and Abolfazl Razi. Timemil: Advancing multivariate time series classification via a time-aware multiple instance learning. *arXiv preprint arXiv:2405.03140*, 2024.
- [4] Ze Liu, Jia Ning, Yue Cao, Yixuan Wei, Zheng Zhang, Stephen Lin, and Han Hu. Video swin transformer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3202–3211, 2022.
- [5] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11976–11986, 2022.
- [6] Zhengzhong Tu, Hossein Talebi, Han Zhang, Feng Yang, Peyman Milanfar, Alan Bovik, and Yinxiao Li. Maxvit: Multi-axis vision transformer. In *European conference on computer vision*, pages 459–479. Springer, 2022.

**NeuroGenesis: LLM-Guided Dataset and Model Selection**

- 1. Dataset Selection:** Identify or synthesize a toy dataset  $D^*$  based on task  $T$ . Use an existing suitable toy dataset if available; otherwise, generate a synthetic dataset tailored to the task.
- 2. Toy Training:** Train each candidate model  $m \in M$  on the selected dataset  $D^*$ . Compute metrics such as training and test losses  $(L_e, L_t)$ , MAE, and MAPE.
- 3. Model Selection:** Evaluate all trained models and perform statistical significance testing with threshold  $\alpha$ . Select the model  $M^*$  demonstrating the highest stability and accuracy using LLM.
- 4. Model Optimization:** Refine  $M^*$  using LLM-guided architectural tweaks and hyperparameter tuning (e.g., learning rate  $\eta$ , dropout  $d$ , batch size  $b$ ). If initially trained on toy data, re-train on the full-scale dataset.

Figure 1: NeuroGenesis algorithm: LLM-guided pipeline for selecting toy datasets, identifying optimal models, and applying model refinement.